Final frozen-model decision

Two locked cohorts • unchanged models and thresholds • paired statistics

Retain weighted HYB-MS real-only as the conservative final reference

The +10% model passed the 30K primary rule but increased FPR on the separate long-URL stress set, so the frozen two-part selection rule does not permit selecting it as the unconditional final model.

What the evidence says

- +10% GAN is better on the large ordinary low-prevalence cohort.
- Real-only produces fewer false alerts on deliberately long legitimate URLs.
- The complete strict selection rule therefore does not yield an unconditional GAN winner.

Deployment status: NOT READY — operational browser validation is still required.

Primary confirmation

30,000 URLs • 5% phishing • zero URL/host/encoding overlap

Condition	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR	FNR
HYB-MS real-only	97.85%	77.16%	81.07%	79.06%	96.58%	84.29%	1.26%	18.93%
HYB-MS +10% GAN	98.12%	80.95%	81.60%	81.27%	96.82%	87.27%	1.01%	18.40%

PRIMARY AUGMENTATION RULE PASSED

Recall, F1 and ROC-AUC improved without a higher FPR.

All rows: McNemar p = 7.427e-05

Legitimate rows: p = 1.863e-05 Phishing rows: p = 0.5264

Bootstrap intervals and all checkpoint/dataset hashes are stored in the machine-readable result.

Long-URL stress test

2,000 long legitimate paths/queries + 200 phishing URLs

Condition	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR	FNR
HYB-MS real-only	96.95%	88.00%	77.00%	82.13%	98.51%	92.45%	1.05%	23.00%
HYB-MS +10% GAN	96.86%	84.66%	80.00%	82.26%	98.42%	91.39%	1.45%	20.00%

LONG-URL NON-REGRESSION FAILED

GAN augmentation increased false positives on long legitimate URLs.

All rows: McNemar p = 0.8714

Legitimate rows: p = 0.07681 Phishing rows: p = 0.2863

Bootstrap intervals and all checkpoint/dataset hashes are stored in the machine-readable result.